15.2.1 Exercises - Molecular kinetic energy

1.

a. State what is meant by the Avogadro constant N_A.

.....

.....[2]

b. A balloon is filled with helium gas at a pressure of 1.1×10^5 Pa and a temperature of 25 ℃. The balloon has a volune of 6.5×10^4 cm^3. Helium may be assumed to be an ideal gas. Determine the number of gas atoms in the balloon.

number = ... [4]

c. Define pressure.

.....[1]

d. Use the kinetic model to explain the pressure exerted by a gas.

.....

.....

.....

.....[4]

e. Explain whether the collisions between the molecules of an ideal gas are elastic or inelastic.

.....

.....[2]

f. State two assumptions of the simple kinetic model of a gas.

.....

.....[2]

g. Use the kinetic model of gases and Newton's laws of motion to explain how a gas exerts a pressure on the sides of its container.

.....

.....

.....[3]

[Total: 18]

2.

a. Some gas, initially at a temperature of 27.2 °C, is heated so that its temperature rises to 38.8 °C. Calculate, in kelvin, to an appropriate number of decimal places,

i. the initial temperature of the gas,

initial temperature = ... K [2]

ii. the rise in temperature

rise in temperature = ... K [1]

b. The pressure p of an ideal gas is given by the expression

$$p = \frac{1}{3}\rho < c^2 >$$

where ρ is the density of the gas.

i. State the meaning of the symbol $<c^2>$.

.....[1]

ii. Use the expression to show that the mean kinetic energy <EK> of the atoms of an ideal gas is given by the expression

$$< E_k >= \frac{1}{3}kT$$

Explain any symbols that you use. [4]

c. Helium-4 may be assumed to behave as an ideal gas.

A cylinder has a constant volume of 7.8×10^3 cm^3 and contains helium-4 gas at a pressure of 2.1×10^7 Pa and at a temperature of 290 K.

Calculate, for the helium gas,

i. the amount of gas,

amount = ... mol [2]

ii. the mean kinetic energy of the atoms,

mean kinetic energy = ... J [2]

iii. the total internal energy.

total kinetic energy = ... J [2]

[Total: 14]

3.

a. Use one of the assumptions of the kinetic theory of gases to explain why the potential energy of the molecules of an ideal gas is zero.aa

.....

.....[1]

b. The average translational kinetic energy E_K of a molecule of an ideal gas is given by the expression

$$E_k = \frac{1}{2}m < c^2 >= \frac{3}{2}kT$$

where m is the mass of a molecule and k is the Boltzmann constant.

State the meaning of the symbol

i. $<c^2>$,

.....[1]

ii. T.

.....[1]

c. A cylinder of constant volume 4.7×10^4 cm^3 contains an ideal gas at pressure 2.6×10^5Pa and temperature 173°C.

The gas is heated. The thermal energy transferred to the gas is 2900J. The final temperature and pressure of the gas are T and p, as illustrated in Fig.

i. Calculate

a) the number N of molecules in the cylinder,

N = ...[3]

b) the increase in average kinetic energy of a molecule during the heating process.

Increase = ...[1]

ii. Use your answer in (i) part b) to determine the final temperature T, in kelvin, of the gas in the cylinder.

T = ...K[3]

[Total: 10]